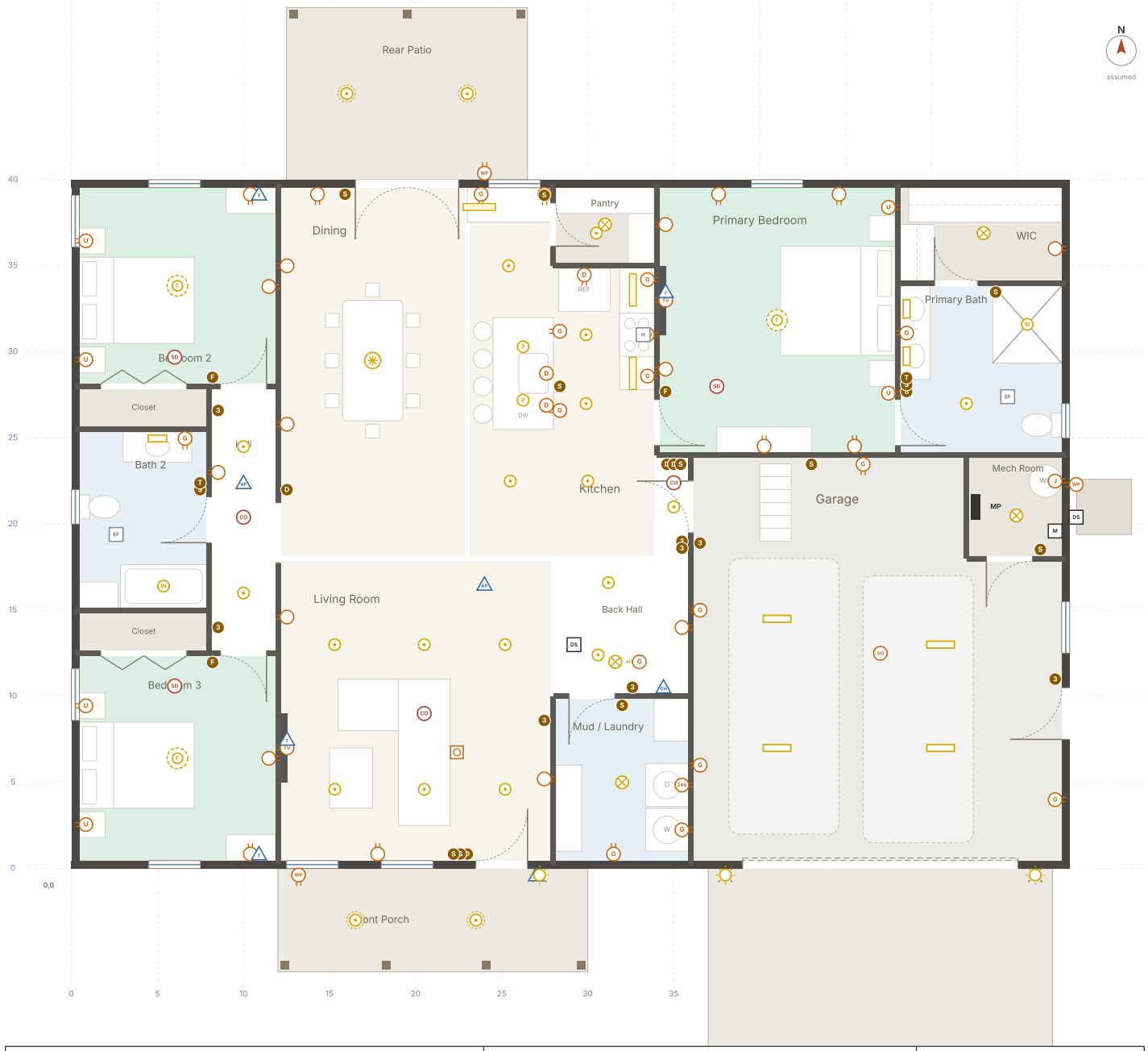


Waring Build

BuildWise plan 3933 · 3 bed, 2 bath · 1,800 sq ft + 528 sq ft garage · 58'-0" x 40'-0" framing
Slab on grade · 9' ceilings · 6:12 gable · grid 5', 0,0 at the south-west corner

Electrical plan

Receptacles, switches, lights, alarms, data; circuits from panel MP
Schematic for pricing and coordination, not a stamped construction document. Verify on site.

E-1

rev
drawn for K. Christianson

E-1 Electrical

receptacles R# switches S# lights and fans L# alarms A# data and low voltage T# panel and disconnects X#

Notes

1. Positions are taped from the nearest wall framing: interior walls to the face of the studs, exterior walls to the inside face of the exterior studs. W# is the wall number on S-1. Ceiling items give two distances, one to each nearest wall.
2. Code basis: NEC 2023 (confirm the edition the county has adopted). AFCI on every 120V 15A and 20A branch circuit in dwelling rooms (210.12); GFCI in bathrooms, kitchen, laundry, garage, outdoors, and for the dishwasher (210.8). Tamper-resistant receptacles throughout (406.12).
3. Mounting heights: receptacles 16" AFF to centre; switches 48"; counter and vanity receptacles 42"–44"; TV boxes 54". Every receptacle position meets the 6'/12' rule along the wall line (210.52(A)); kitchen counters have no point more than 24" from a receptacle (210.52(C)).
4. Panel MP is in the garage mechanical room on wall 20, 200A main breaker. Service entrance from the meter on the east wall; grounding electrode: Ufer in the footing plus a ground rod.
5. Smoke alarms in every bedroom, smoke/CO alarms outside the bedrooms and at the garage entry, all hardwired, interconnected, with battery backup (R314, R315).
6. Circuit sheets E-1.1 onward show one circuit each with its cable run drawn as a chain from MP through every device in nearest-neighbour order (solid for receptacle and appliance circuits, dashed for lighting). The run is a proposed order, not a required one. E-1.D shows the Cat6 home runs to the media enclosure.
7. Switch plan: every light belongs to one switch leg (table below). Clicking a light lights up every device on its leg in blue; clicking a leg row does the same.
8. All-electric house assumed: electric range, electric water heater, heat pump with electric strip backup. NEC 220.82 estimate: about 32 kVA, 135A, inside the 200A service.

Open items

The particulars to settle on this sheet.

- All-electric assumed. If natural gas or propane is coming, the range, water heater and backup heat change, and circuits 1-2, 5-6 and 9-10 shrink.
- EV charging: add a 240V 50A circuit to the garage now (about $350 rough-in), or leave a spare 2-pole space and conduit?
- Ceiling fans in the three bedrooms drawn on fan-rated boxes. Add one in the living room?
- Data: Cat6 to each bedroom, both TVs and two ceiling access points drawn. Add cameras at the doors or more drops?
- Heat pump and air handler breakers are placeholders until the HVAC contractor picks the equipment (nameplate MCA / MOCP).

Circuit schedule, panel MP

MP: 200A main breaker load centre, 40 spaces, in the garage mechanical room. Circuit numbers are breaker positions; 2-pole loads take two.

Ckt	Load	Breaker	Wire	Serves	Devices
1-2	Range	240V 50A 2P	#6 Cu NM	kitchen, wall 15	1
3-4	Dryer	240V 30A 2P	#10 Cu NM (4-wire)	mud / laundry	1
5-6	Water heater	240V 30A 2P	#10 Cu NM	mechanical room	1
7-8	Heat pump HP-1 (outdoor unit)	240V 30A 2P (per nameplate MOCP)	#10 Cu	east wall, disconnect at the unit	1
9-10	Air handler AH-1 with 10 kW heat strips	240V 60A 2P (per nameplate MOCP)	#6 Cu	attic, disconnect at the unit	1
11	Kitchen small appliance 1	20A AFCI/GFCI	#12 Cu	kitchen counters: rear run, wall run south	3
12	Kitchen small appliance 2	20A AFCI/GFCI	#12 Cu	kitchen counters: wall run north, island	3
13	Dishwasher	20A AFCI/GFCI	#12 Cu	island	1
14	Disposal	20A AFCI/GFCI	#12 Cu	island	2
15	Refrigerator	20A AFCI	#12 Cu	kitchen	1
16	Laundry	20A AFCI/GFCI	#12 Cu	washer and laundry receptacles (210.11(C)(2))	2

Ckt	Load	Breaker	Wire	Serves	Devices
17	Bath 2 receptacles	20A GFCI	#12 Cu	Bath 2 (210.11(C)(3))	1
18	Primary bath receptacles	20A GFCI	#12 Cu	primary bath (210.11(C)(3))	1
19	Bedroom 3 and hall receptacles	15A AFCI	#14 Cu	Bedroom 3, hall	5
20	Bedroom 2 receptacles	15A AFCI	#14 Cu	Bedroom 2	4
21	Primary bedroom and WIC receptacles	15A AFCI	#14 Cu	primary bedroom, WIC	10
22	Living room receptacles	15A AFCI	#14 Cu	living room	5
23	Dining and back hall receptacles	20A AFCI	#12 Cu	dining, back hall	4
24	Lighting, bedroom wing	15A AFCI	#14 Cu	Bedrooms 2 and 3, hall, Bath 2 lights and fan, smoke alarms	19
25	Lighting, great room	15A AFCI	#14 Cu	living, dining, kitchen, pantry, back hall, laundry, porch and patio	41
26	Lighting, primary suite and garage	15A AFCI	#14 Cu	primary suite, garage, mechanical room, attic, garage coach lights	26
27	Garage receptacles	20A GFCI	#12 Cu	garage, attic service receptacle (210.11(C)(4), 210.63)	5
28	Garage door opener	20A GFCI	#12 Cu	opener receptacle in the garage ceiling	1
29	Exterior receptacles	20A GFCI	#12 Cu	porch, patio, HP-1 service receptacle (210.63)	3

Switch plan

One row per switch leg. Click a row to light up its switch and lights on the drawing.

Leg	What	Room	Switch(es)	Lights	Control
mech	Mech room light	Mech Room	S1 single-pole switch, rocker	L1 surface LED ceiling fixture	switch inside the door
bed3	Bedroom 3 fan and light	Bedroom 3	S2 fan speed control and light switch (2-gang)	L2 fan-rated ceiling box	fan speed and light at D6
bed2	Bedroom 2 fan and light	Bedroom 2	S3 fan speed control and light switch (2-gang)	L3 fan-rated ceiling box	fan speed and light at D10
hall	Hall cans	Hall	S4 3-way switch, rocker; S5 3-way switch, rocker	L4 6" recessed LED wafer, L5 6" recessed LED wafer	3-way: each end of the hall
bath2	Bath 2 lights	Bath 2	S6 single-pole switch, rocker	L6 vanity light bar over the mirror, L7 6" recessed LED wafer	vanity light and tub can on one switch at D8
bath2-fan	Bath 2 fan	Bath 2	S7 timer switch for the exhaust fan	L8 bath exhaust fan	timer at D8
living	Living room cans	Living Room	S8 dimmer, rocker, LED rated; S11 3-way switch, rocker	L9 6" recessed LED wafer, L10 6" recessed LED wafer, L11 6" recessed LED wafer, L12 6" recessed LED wafer, L13 6" recessed LED wafer, L14 6" recessed LED wafer	3-way dimmer: front door and the kitchen side of wall 11
porch	Porch lights	Front Porch	S9 single-pole switch, rocker	L15 exterior soffit downlight, L16 exterior soffit downlight, L17 exterior wall light	at the front door
coach	Garage coach lights	Exterior	S10 single-pole switch, rocker	L20 exterior wall light, L21 exterior wall light	at the front door; dusk-to-dawn sensors in the fixtures
patio	Patio lights	Rear Patio	S13 single-pole switch, rocker	L18 exterior soffit downlight, L19 exterior soffit downlight	beside D2
dining	Dining chandelier	Dining	S12 dimmer, rocker, LED rated	L22 chandelier box	dimmer on wall 5
kitchen	Kitchen cans	Kitchen	S14 dimmer, rocker, LED rated	L29 6" recessed LED wafer, L30 6" recessed LED wafer, L31 6" recessed LED wafer, L32 6" recessed LED wafer, L33 6" recessed LED wafer, L34 6" recessed LED wafer	dimmer on wall 14, back-hall alcove

Leg	What	Room	Switch(es)	Lights	Control
pendants	Island pendants	Kitchen	S15 dimmer, rocker, LED rated	L27 pendant over the island, L28 pendant over the island	dimmer on wall 14
undercab	Under-cabinet lights	Kitchen	S16 single-pole switch, rocker	L24 under-cabinet LED strip with driver, L25 under-cabinet LED strip with driver, L26 under-cabinet LED strip with driver	switch on wall 14
hood	Range hood	Kitchen	—	L23 range hood	controls on the hood
pantry	Pantry light	Pantry	S18 single-pole switch, rocker	L35 surface LED ceiling fixture	kitchen side of D15
backhall	Back hall cans	Back Hall	S19 3-way switch, rocker; S21 3-way switch, rocker	L36 6" recessed LED wafer, L37 6" recessed LED wafer, L38 6" recessed LED wafer	3-way: D3 and D11
laundry	Laundry light	Mud / Laundry	S22 single-pole switch, rocker	L39 surface LED ceiling fixture	at D11
primary	Primary bedroom fan and light	Primary Bedroom	S23 fan speed control and light switch (2-gang)	L40 fan-rated ceiling box	fan speed and light at D12
pbath	Primary bath vanity lights	Primary Bath	S24 single-pole switch, rocker	L41 vanity light bar over the mirror, L42 vanity light bar over the mirror	at D13
pbath-can	Primary bath cans	Primary Bath	S25 single-pole switch, rocker	L43 6" recessed LED wafer, L44 6" recessed LED wafer	at D13
pbath-fan	Primary bath fan	Primary Bath	S26 timer switch for the exhaust fan	L45 bath exhaust fan	continuous low with a boost timer at D13
wic	WIC light	WIC	S27 single-pole switch, rocker	L46 surface LED ceiling fixture	bath side of D14
garage	Garage lights	Garage	S20 3-way switch, rocker; S28 3-way switch, rocker; S29 3-way switch, rocker	L47 4' LED shop light, L48 4' LED shop light, L49 4' LED shop light, L50 4' LED shop light	3-way: D3 (house side and garage side) and D5
attic	Attic light	Attic	S30 single-pole switch, rocker	L51 attic light at the air handler	switch with a pilot light by the attic stair

Device schedule

Tag	Device	Position	AFF	Circuit	Notes
X1	main panel MP, 200A main breaker, 40 space	on W20, east face, 2'-10" from W14	60"	—	Mech Room. Main panel MP on wall 20, garage-side mech room; 30" x 36" working space in front (110.26), service entrance from the meter on the east wall
X2	structured media enclosure, 28"	on the east ext. wall, 1'-5" from W21	60"	—	Mech Room. Structured media enclosure on wall 2: every Cat6 home run lands here; doorbell transformer beside it
R1	junction box, hardwired appliance	on the east ext. wall, 1'-4" from W14	48"	5-6	Mech Room. Water heater: hardwired from a box on wall 2, disconnect within sight
L1	surface LED ceiling fixture, 3000K	2'-8" from east ext. wall, 2'-4" from W21	108"	26	Mech Room
S1	single-pole switch, rocker	on W21, north face, 1'-3" from east ext. wall	48"	26	Mech Room. inside, latch side of D16
X3	AC disconnect, fused / non-fused per nameplate	outside, 58'-0" E, 20'-5" N of the datum	48"	7-8	Exterior. HP-1 disconnect on the east wall, within sight of the unit
R2	GFCI receptacle, 20A, weather resistant, in-use cover	outside, 58'-0" E, 22'-4" N of the datum	24"	29	Exterior. service receptacle within 25' of HP-1 (210.63)
R3	duplex receptacle, 15A, tamper resistant, with USB-A/C	on the west ext. wall, 2'-1" from front ext. wall	16"	19	Bedroom 3. nightstand, south side of the bed

Tag	Device	Position	AFF	Circuit	Notes
R4	duplex receptacle, 15A, tamper resistant, with USB-A/C	on the west ext. wall, 2'-11" from W6	16"	19	Bedroom 3. nightstand, north side of the bed
R5	duplex receptacle, 15A, tamper resistant	on W5, west face, 5'-11" from front ext. wall	16"	19	Bedroom 3
R6	duplex receptacle, 15A, tamper resistant	on the front ext. wall, 1'-5" from W5	16"	19	Bedroom 3. behind the dresser, for a TV on it
T1	Cat6 data outlet	on the front ext. wall, 0'-11" from W5	16"	—	Bedroom 3
L2	fan-rated ceiling box, ceiling fan with light kit	5'-8" from W5, 5'-11" from front ext. wall	108"	24	Bedroom 3. centre of the room
S2	fan speed control and light switch (2-gang)	on W6, south face, 3'-8" from W5	48"	24	Bedroom 3. latch side of D6
A1	smoke alarm, hardwired, interconnected, battery backup	5'-7" from west ext. wall, 1'-9" from W6	108"	24	Bedroom 3
R7	duplex receptacle, 15A, tamper resistant, with USB-A/C	on the west ext. wall, 1'-5" from W10	16"	20	Bedroom 2. nightstand, south side of the bed
R8	duplex receptacle, 15A, tamper resistant, with USB-A/C	on the west ext. wall, 3'-1" from rear ext. wall	16"	20	Bedroom 2. nightstand, north side of the bed
R9	duplex receptacle, 15A, tamper resistant	on W5, west face, 5'-8" from W10	16"	20	Bedroom 2
R10	duplex receptacle, 15A, tamper resistant	on the rear ext. wall, 1'-5" from W5	16"	20	Bedroom 2. behind the dresser, for a TV on it
T2	Cat6 data outlet	on the rear ext. wall, 0'-11" from W5	16"	—	Bedroom 2
L3	fan-rated ceiling box, ceiling fan with light kit	5'-8" from W5, 5'-8" from W10	108"	24	Bedroom 2. centre of the room
S3	fan speed control and light switch (2-gang)	on W10, north face, 3'-8" from W5	48"	24	Bedroom 2. latch side of D10
A2	smoke alarm, hardwired, interconnected, battery backup	5'-7" from west ext. wall, 1'-7" from W10	108"	24	Bedroom 2
L4	6" recessed LED wafer, 3000K, IC	1'-10" from W5, 3'-4" from W6	108"	24	Hall
L5	6" recessed LED wafer, 3000K, IC	1'-10" from W5, 3'-4" from W10	108"	24	Hall
S4	3-way switch, rocker	on W8, east face, 1'-4" from W6	48"	24	Hall. south end, by D6
S5	3-way switch, rocker	on W8, east face, 1'-3" from W10	48"	24	Hall. north end, by D10
R11	duplex receptacle, 15A, tamper resistant	on W8, east face, 4'-10" from W10	16"	19	Hall. hall over 10' long (210.52(H))
A3	smoke / CO alarm, hardwired, interconnected, battery backup	1'-10" from W5, 7'-5" from W10	108"	24	Hall. outside the bedrooms
T3	Cat6 drop for a ceiling wireless access point	1'-10" from W5, 5'-5" from W10	108"	—	Hall. wireless access point, bedroom wing

Tag	Device	Position	AFF	Circuit	Notes
R12	GFCI receptacle, 20A, tamper resistant	on W9, south face, 1'-3" from W8	42"	17	Bath 2. above the vanity backsplash, within 3' of the basin
L6	vanity light bar over the mirror	on W9, south face, 2'-10" from W8	84"	24	Bath 2. over the mirror
L7	6" recessed LED wafer, 3000K, IC, wet location	2'-6" from W8, 1'-3" from W7	108"	24	Bath 2. over the tub
L8	bath exhaust fan, 80 CFM, 0.8 sone, ducted to a roof cap	2'-2" from west ext. wall, 4'-3" from W7	108"	24	Bath 2. between the toilet and the tub
S6	single-pole switch, rocker	on W8, west face, 3'-4" from W9	48"	24	Bath 2. latch side of D8: 3-gang with the next two
S7	timer switch for the exhaust fan	on W8, west face, 2'-11" from W9	48"	24	Bath 2. fan timer
R13	duplex receptacle, 15A, tamper resistant	on the front ext. wall, 5'-8" from W5	16"	22	Living Room
R14	TV box: duplex receptacle and low-voltage pass-through	on W5, east face, 6'-7" from front ext. wall	54"	22	Living Room. behind the TV on wall 5
T4	Cat6 data outlet	on W5, east face, 7'-1" from front ext. wall	54"	—	Living Room. behind the TV: 2 x Cat6
R15	duplex receptacle, 15A, tamper resistant	on W5, east face, 14'-2" from front ext. wall	16"	22	Living Room
R16	duplex receptacle, 15A, tamper resistant	on W11, west face, 4'-9" from front ext. wall	16"	22	Living Room
R17	floor box with duplex receptacle, 15A	5'-5" from W11, 6'-4" from front ext. wall	—	22	Living Room. floor box behind the sectional, for lamps
L9	6" recessed LED wafer, 3000K, IC	3'-2" from W5, 4'-2" from front ext. wall	108"	25	Living Room
L10	6" recessed LED wafer, 3000K, IC	3'-2" from W5, 12'-6" from front ext. wall	108"	25	Living Room
L11	6" recessed LED wafer, 3000K, IC	7'-4" from W11, 4'-2" from front ext. wall	108"	25	Living Room
L12	6" recessed LED wafer, 3000K, IC	8'-4" from W5, 12'-6" from front ext. wall	108"	25	Living Room
L13	6" recessed LED wafer, 3000K, IC	2'-8" from W11, 4'-2" from front ext. wall	108"	25	Living Room
L14	6" recessed LED wafer, 3000K, IC	10'-8" from W13, 12'-6" from front ext. wall	108"	25	Living Room
S8	dimmer, rocker, LED rated	on the front ext. wall, 4'-10" from W11	48"	25	Living Room. front door, latch side of D1: 3-gang (living cans 3-way, porch, garage coach lights)
S9	single-pole switch, rocker	on the front ext. wall, 5'-3" from W11	48"	25	Living Room. front door bank
S10	single-pole switch, rocker	on the front ext. wall, 5'-8" from W11	48"	26	Living Room. front door bank
S11	3-way switch, rocker	on W11, west face, 8'-2" from front ext. wall	48"	25	Living Room. living cans, kitchen side

Tag	Device	Position	AFF	Circuit	Notes
A4	smoke / CO alarm, hardwired, interconnected, battery backup	7'-4" from W11, 8'-6" from front ext. wall	108"	24	Living Room. great room, over 20' from the range
T5	Cat6 drop for a ceiling wireless access point	11'-10" from W13, 16'-1" from front ext. wall	108"	—	Living Room. wireless access point, great room
T6	video doorbell (low voltage, transformer in the mech room)	outside, 27'-0" E, 0'-0" N of the datum	48"	—	Exterior. video doorbell beside D1
L15	exterior soffit downlight, wet location	3'-6" from front ext. wall	108"	25	Front Porch
L16	exterior soffit downlight, wet location	3'-6" from front ext. wall	108"	25	Front Porch
L17	exterior wall light, wet location, dusk-to-dawn	outside, 27'-2" E, 0'-0" N of the datum	72"	25	Front Porch. beside the front door
R18	GFCI receptacle, 20A, weather resistant, in-use cover	outside, 13'-2" E, 0'-0" N of the datum	16"	29	Front Porch
L18	exterior soffit downlight, wet location	5'-5" from rear ext. wall	108"	25	Rear Patio
L19	exterior soffit downlight, wet location	5'-5" from rear ext. wall	108"	25	Rear Patio
R19	GFCI receptacle, 20A, weather resistant, in-use cover	outside, 24'-0" E, 40'-0" N of the datum	16"	29	Rear Patio
L20	exterior wall light, wet location, dusk-to-dawn	outside, 38'-0" E, 0'-0" N of the datum	84"	26	Exterior. coach light, west of the garage door
L21	exterior wall light, wet location, dusk-to-dawn	outside, 56'-0" E, 0'-0" N of the datum	84"	26	Exterior. coach light, east of the garage door
R20	duplex receptacle, 15A, tamper resistant	on W5, east face, 13'-9" from rear ext. wall	16"	23	Dining
R21	duplex receptacle, 15A, tamper resistant	on W5, east face, 4'-7" from rear ext. wall	16"	23	Dining
R22	duplex receptacle, 15A, tamper resistant	on the rear ext. wall, 2'-2" from W5	16"	23	Dining
L22	chandelier box, dimmable	5'-4" from W5, 10'-1" from rear ext. wall	96"	25	Dining. over the table centre
S12	dimmer, rocker, LED rated	on W5, east face, 17'-7" from rear ext. wall	48"	25	Dining. north of the cased opening
S13	single-pole switch, rocker	on the rear ext. wall, 3'-9" from W5	48"	25	Dining. beside D2
R23	GFCI receptacle, 20A, tamper resistant	on the rear ext. wall, 4'-1" from W18	44"	11	Kitchen. rear run
R24	GFCI receptacle, 20A, tamper resistant	on the rear ext. wall, 0'-4" from W18	44"	11	Kitchen. rear run, east of the window
R25	GFCI receptacle, 20A, tamper resistant	on W15, west face, 6'-3" from W19	44"	11	Kitchen. wall run, south of the range
R26	GFCI receptacle, 20A, tamper resistant	on W15, west face, 0'-8" from W19	44"	12	Kitchen. wall run, north of the range
R27	240V receptacle / appliance connection	on W15, west face, 3'-10" from W19	4"	1-2	Kitchen. range receptacle behind the range

Tag	Device	Position	AFF	Circuit	Notes
L23	range hood, 300 CFM, ducted 6" to a roof cap	outside, 33'-7" E, 31'-0" N of the datum	66"	25	Kitchen. range hood under the upper cabinet
R28	GFCI receptacle, 20A, tamper resistant	outside, 28'-0" E, 26'-7" N of the datum	30"	12	Kitchen. island, east face, south end
R29	GFCI receptacle, 20A, tamper resistant	outside, 28'-0" E, 31'-2" N of the datum	30"	12	Kitchen. island, east face, north end
R30	dedicated receptacle, 20A	6'-3" from W15, 12'-8" from rear ext. wall	6"	13	Kitchen. dishwasher, in the sink base
R31	dedicated receptacle, 20A	6'-3" from W15, 10'-10" from rear ext. wall	12"	14	Kitchen. disposal, switched at the sink
R32	dedicated receptacle, 20A	on W19, south face, 4'-1" from W15	48"	15	Kitchen. behind the refrigerator
L24	under-cabinet LED strip with driver	outside, 33'-0" E, 28'-9" N of the datum	54"	25	Kitchen. under the wall-run uppers, south
L25	under-cabinet LED strip with driver	outside, 33'-0" E, 33'-7" N of the datum	54"	25	Kitchen. under the wall-run uppers, north
L26	under-cabinet LED strip with driver	outside, 23'-8" E, 38'-10" N of the datum	54"	25	Kitchen. under the rear-run uppers
L27	pendant over the island	7'-7" from W15, 12'-4" from rear ext. wall	78"	25	Kitchen. over the island
L28	pendant over the island	7'-7" from W15, 9'-3" from rear ext. wall	78"	25	Kitchen. over the island
L29	6" recessed LED wafer, 3000K, IC	5'-10" from W13, 12'-4" from W19	108"	25	Kitchen
L30	6" recessed LED wafer, 3000K, IC	10'-4" from W13, 17'-1" from rear ext. wall	108"	25	Kitchen
L31	6" recessed LED wafer, 3000K, IC	3'-11" from W15, 7'-10" from W19	108"	25	Kitchen
L32	6" recessed LED wafer, 3000K, IC	3'-11" from W15, 3'-10" from W19	108"	25	Kitchen
L33	6" recessed LED wafer, 3000K, IC	2'-5" from W18, 4'-7" from rear ext. wall	108"	25	Kitchen
L34	6" recessed LED wafer, 3000K, IC	2'-4" from W18, 1'-9" from W19	108"	25	Kitchen
S14	dimmer, rocker, LED rated	on W14, south face, 1'-3" from W13	48"	25	Kitchen. kitchen bank on wall 14 in the back-hall alcove: 4-gang
S15	dimmer, rocker, LED rated	on W14, south face, 0'-10" from W13	48"	25	Kitchen. kitchen bank
S16	single-pole switch, rocker	on W14, south face, 0'-5" from W13	48"	25	Kitchen. kitchen bank
S17	single-pole switch, rocker	outside, 28'-0" E, 28'-0" N of the datum	36"	14	Kitchen. disposal switch in the island face, beside the sink
L35	surface LED ceiling fixture, 3000K	2'-10" from W15, 2'-2" from rear ext. wall	108"	25	Pantry

Tag	Device	Position	AFF	Circuit	Notes
S18	single-pole switch, rocker	on W18, west face, 0'-5" from rear ext. wall	48"	25	Pantry. kitchen side, latch side of D15
L36	6" recessed LED wafer, 3000K, IC	5'-3" from W13, 2'-3" from W12	108"	25	Back Hall
L37	6" recessed LED wafer, 3000K, IC	4'-8" from W13, 6'-5" from W12	108"	25	Back Hall
L38	6" recessed LED wafer, 3000K, IC	0'-10" from W13, 2'-10" from W14	108"	25	Back Hall
S19	3-way switch, rocker	on W13, west face, 4'-10" from W14	48"	25	Back Hall. latch side of D3: 2-gang with the garage lights
S20	3-way switch, rocker	on W13, west face, 5'-3" from W14	48"	26	Back Hall. garage lights, house side
S21	3-way switch, rocker	on W12, north face, 3'-3" from W13	48"	25	Back Hall. by D11
R33	duplex receptacle, 15A, tamper resistant	on W13, west face, 3'-10" from W12	16"	23	Back Hall
A5	smoke / CO alarm, hardwired, interconnected, battery backup	0'-10" from W13, 1'-5" from W14	108"	24	Back Hall. by the garage door and the primary suite
T7	door chime	on W12, north face, 1'-5" from W13	84"	—	Back Hall
R34	GFCI receptacle, 20A, tamper resistant	on W13, west face, 1'-10" from front ext. wall	42"	16	Mud / Laundry. washer, beside the washer box
R35	240V receptacle / appliance connection	on W13, west face, 4'-5" from front ext. wall	30"	3-4	Mud / Laundry. dryer, NEMA 14-30R
R36	GFCI receptacle, 20A, tamper resistant	on the front ext. wall, 3'-4" from W11	42"	16	Mud / Laundry. over the bench
L39	surface LED ceiling fixture, 3000K	3'-10" from W13, 4'-7" from front ext. wall	108"	25	Mud / Laundry
S22	single-pole switch, rocker	on W12, south face, 3'-10" from W13	48"	25	Mud / Laundry. latch side of D11
R37	duplex receptacle, 15A, tamper resistant, with USB-A/C	on W16, west face, 3'-5" from W14	16"	21	Primary Bedroom. nightstand, south side of the bed
R38	duplex receptacle, 15A, tamper resistant, with USB-A/C	on W16, west face, 1'-2" from rear ext. wall	16"	21	Primary Bedroom. nightstand, north side of the bed
R39	TV box: duplex receptacle and low-voltage pass-through	on W15, east face, 6'-7" from rear ext. wall	54"	21	Primary Bedroom. behind the TV on wall 15
T8	Cat6 data outlet	on W15, east face, 6'-1" from rear ext. wall	54"	—	Primary Bedroom. behind the TV
R40	duplex receptacle, 15A, tamper resistant	on W15, east face, 4'-10" from W14	16"	21	Primary Bedroom
R41	duplex receptacle, 15A, tamper resistant	on W15, east face, 2'-2" from rear ext. wall	16"	21	Primary Bedroom
R42	duplex receptacle, 15A, tamper resistant	on the rear ext. wall, 3'-5" from W15	16"	21	Primary Bedroom

Tag	Device	Position	AFF	Circuit	Notes
R43	duplex receptacle, 15A, tamper resistant	on the rear ext. wall, 3'-3" from W16	16"	21	Primary Bedroom
R44	duplex receptacle, 15A, tamper resistant	on W14, north face, 6'-1" from W15	16"	21	Primary Bedroom. behind the dresser
R45	duplex receptacle, 15A, tamper resistant	on W14, north face, 2'-4" from W16	16"	21	Primary Bedroom
L40	fan-rated ceiling box, ceiling fan with light kit	6'-10" from W15, 7'-8" from W14	108"	26	Primary Bedroom. centre of the room
S23	fan speed control and light switch (2-gang)	on W15, east face, 3'-7" from W14	48"	26	Primary Bedroom. latch side of D12
A6	smoke alarm, hardwired, interconnected, battery backup	3'-4" from W15, 3'-10" from W14	108"	24	Primary Bedroom
R46	GFCI receptacle, 20A, tamper resistant	on W16, east face, 2'-9" from W17	42"	18	Primary Bath. between the basins, above the backsplash
L41	vanity light bar over the mirror	on W16, east face, 4'-1" from W17	84"	26	Primary Bath. over the south basin
L42	vanity light bar over the mirror	on W16, east face, 1'-4" from W17	84"	26	Primary Bath. over the north basin
L43	6" recessed LED wafer, 3000K, IC, wet location	2'-0" from east ext. wall, 2'-3" from W17	108"	26	Primary Bath. in the shower
L44	6" recessed LED wafer, 3000K, IC	3'-10" from W16, 2'-10" from W14	108"	26	Primary Bath
L45	bath exhaust fan, 80 CFM, 0.8 sone, ducted to a roof cap	3'-2" from east ext. wall, 3'-3" from W14	108"	26	Primary Bath. runs continuous at 50 CFM for whole-house ventilation (M1505.4), boost to 80
S24	single-pole switch, rocker	on W16, east face, 3'-7" from W14	48"	26	Primary Bath. latch side of D13: 3-gang
S25	single-pole switch, rocker	on W16, east face, 3'-11" from W14	48"	26	Primary Bath. 3-gang
S26	timer switch for the exhaust fan	on W16, east face, 4'-4" from W14	48"	26	Primary Bath. fan control: continuous low, boost timer
L46	surface LED ceiling fixture, 3000K	4'-7" from east ext. wall, 2'-8" from rear ext. wall	108"	26	WIC
S27	single-pole switch, rocker	on W17, south face, 3'-10" from east ext. wall	48"	26	WIC. bath side, latch side of D14
R47	duplex receptacle, 15A, tamper resistant	on the east ext. wall, 1'-10" from W17	16"	21	WIC
R48	GFCI receptacle, 20A, tamper resistant	on W13, east face, 5'-7" from front ext. wall	48"	27	Garage
R49	GFCI receptacle, 20A, tamper resistant	on W13, east face, 8'-10" from W14	48"	27	Garage
R50	GFCI receptacle, 20A, tamper resistant	on W14, south face, 5'-10" from W20	48"	27	Garage
R51	GFCI receptacle, 20A, tamper resistant	on the east ext. wall, 3'-6" from front ext. wall	48"	27	Garage

Tag	Device	Position	AFF	Circuit	Notes
R52	ceiling receptacle for the garage door opener, 20A GFCI	10'-7" from east ext. wall, 11'-4" from W14	108"	28	Garage. opener receptacle; push button by D3, safety eyes at the door
L47	4' LED shop light, 4000K	4'-10" from W13, 6'-7" from front ext. wall	108"	26	Garage
L48	4' LED shop light, 4000K	4'-10" from W13, 9'-4" from W14	108"	26	Garage
L49	4' LED shop light, 4000K	7'-1" from east ext. wall, 6'-7" from front ext. wall	108"	26	Garage
L50	4' LED shop light, 4000K	7'-1" from east ext. wall, 10'-10" from W14	108"	26	Garage
S28	3-way switch, rocker	on W13, east face, 4'-11" from W14	48"	26	Garage. garage side of D3
S29	3-way switch, rocker	on the east ext. wall, 6'-10" from W21	48"	26	Garage. latch side of D5
S30	single-pole switch, rocker	on W14, south face, 6'-10" from W13	54"	26	Garage. by the attic stair, with a pilot light
L51	attic light at the air handler, switched at the access with a pilot light (M1305.1.3.1)	4'-3" from W13, 1'-10" from W12	120"	26	Attic. at AH-1, on the service platform
R53	GFCI receptacle, 20A, tamper resistant	2'-10" from W13, 1'-10" from W12	120"	27	Attic. service receptacle at AH-1 (210.63)
X4	AC disconnect, fused / non-fused per nameplate	6'-8" from W13, 2'-10" from W12	120"	9-10	Attic. AH-1 disconnect at the unit

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The contractor takes off the final list.

Item	Qty	Notes
duplex receptacle, 15A, tamper resistant	19	
6" recessed LED wafer, 3000K, IC	18	
GFCI receptacle, 20A, tamper resistant	15	
single-pole switch, rocker	13	
3-way switch, rocker	8	
duplex receptacle, 15A, tamper resistant, with USB-A/C	6	
surface LED ceiling fixture, 3000K	4	
Cat6 data outlet	4	
dimmer, rocker, LED rated	4	
exterior soffit downlight, wet location	4	
4' LED shop light, 4000K	4	
GFCI receptacle, 20A, weather resistant, in-use cover	3	
fan-rated ceiling box, ceiling fan with light kit	3	
fan speed control and light switch (2-gang)	3	

Item	Qty	Notes
smoke alarm, hardwired, interconnected, battery backup	3	
smoke / CO alarm, hardwired, interconnected, battery backup	3	
vanity light bar over the mirror	3	
exterior wall light, wet location, dusk-to-dawn	3	
dedicated receptacle, 20A	3	
under-cabinet LED strip with driver	3	
AC disconnect, fused / non-fused per nameplate	2	
Cat6 drop for a ceiling wireless access point	2	
6" recessed LED wafer, 3000K, IC, wet location	2	
bath exhaust fan, 80 CFM, 0.8 sone, ducted to a roof cap	2	
timer switch for the exhaust fan	2	
TV box: duplex receptacle and low-voltage pass-through	2	
240V receptacle / appliance connection	2	
pendant over the island	2	
main panel MP, 200A main breaker, 40 space	1	
structured media enclosure, 28"	1	
junction box, hardwired appliance	1	
floor box with duplex receptacle, 15A	1	
video doorbell (low voltage, transformer in the mech room)	1	
chandelier box, dimmable	1	
range hood, 300 CFM, ducted 6" to a roof cap	1	
door chime	1	
ceiling receptacle for the garage door opener, 20A GFCI	1	
attic light at the air handler, switched at the access with a pilot light (M1305.1.3.1)	1	
14/2 NM-B (and 14/3 for 3-ways)	9 × 250' rolls	about 2152' from the runs on the circuit sheets, +15%
12/2 NM-B	4 × 250' rolls	about 893'
6/3 NM-B (range), 6/2 + ground Cu (AH-1), 10/3 (dryer), 10/2 (water heater, HP-1)	by the run	measure on site
Device boxes, new-work, with covers	87	fan-rated pancake boxes at the fan locations
Cat6 CMR	1 × 1000' box	home runs to the media enclosure